

What Gives You Away?

How Linguistic Cues Move User Readouts Across Three Language Models

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Abstract

Small differences in wording may alter the demographic signals encoded in a language model’s representation of a user. We test this possibility by placing controlled linguistic edits inside otherwise identical conversations. The main experiment compares 1,025 pairs built from 100 base messages in Llama 3.2 3B. A shared set of 505 pairs is also tested in Llama 3.1 8B and OLMo 2 32B. Simple classifiers act as measuring tools: they show how much each change moves internal signals related to the user’s age, gender, education, or socioeconomic status.

In the main experiment, 30 of the 36 combinations of writing cue and user attribute produce a statistically reliable shift after accounting for the number of tests; 28 move more than a matched synonym edit. The age readout is especially sensitive to personal context, emoji, and slang. The education readout is most affected by grammar and personal context, while the socioeconomic readout is most affected by price language and personal context. The gender pattern is less consistent. The broad ordering appears again in both larger models and under different conversation wording. Some style effects fade after later messages, but the effect of personal context lasts longer.

These results show that small differences in wording can change the demographic signals measured inside a model. They do not show that those signals are correct, that the model has identified a person’s real attributes, or that it uses this information when writing its response.

1 Introduction

Writing has long been known to carry information about its author (Stamatatos, 2009; Schler et al., 2006). In a dialogue system, a related question is whether a controlled change to one user turn alters a demographic probe output while the rest of the conversation remains unchanged. This isolates sensitivity to a linguistic cue from demographic prediction over an uncontrolled corpus.

This study uses the age, gender, education, and socioeconomic readouts from TalkTuner (Chen et al., 2024). A linear probe converts a hidden state into class logits, providing a fixed instrument for measuring how an edit changes the representation. A separate five-seed probe ensemble is fitted for each of three models. Rows sharing a source generator and numeric index remain together during training and evaluation, so related source rows cannot cross partitions.

The probes were trained on full conversations, so every primary measurement is also made in a full conversation. Two independently worded neutral preludes place the edited message at the current user turn. Two further conditions move that message one or two user turns into the past. The paired conversations always share the same assistant messages, preventing a generated response from becoming an uncontrolled mediator.

The experiment maps nine cue categories to four readouts, evaluates that map in three checkpoints, and measures selected effects again after one and two later user turns. It also varies the neutral prelude, the readout phrase, and the probe split seed. Cue order is similar across models, with greater variation in lower-ranked cues and temporal persistence.

2 Related work

User representations. TalkTuner trains linear probes for a conversational user’s age, gender, education, and socioeconomic status, then exposes those estimates in an interactive dashboard (Chen et al., 2024). Related work examines implicit personalization and stereotype-driven user representations (Neplenbroek et al., 2025). This study measures the sensitivity of those readouts to controlled input edits.

Probing and interpretation. Probe performance can reflect both information in a representation and properties of the probe, while intervention-based work separately tests whether encoded information affects model behavior (Hewitt and Liang, 2019; Voita and Titov, 2020; Elazar et al., 2021). Our paired contrast identifies the effect of a particular text edit on a fixed probe output in a constructed conversation. It says nothing by itself about label truth or downstream generation.

Prompt sensitivity is a related measurement problem: small changes to an evaluation prompt can alter model scores without changing the underlying task (Hua et al., 2025). Sensitivity to readout phrasing is evaluated by repeating the cue matrix under three paraphrases of each attribute’s readout phrase.

Style and demographic inference. Stylometric systems use vocabulary, syntax, spelling, and punctuation to infer properties of authors and documents (Stamatatos, 2009). Such correlations can reflect population structure, topic, sampling choices, and stereotypes. A paired design isolates changes in probe output while the request and conversation are held fixed.

3 Method

3.1 Model and readouts

The main experiment uses Llama 3.2 3B Instruct (Meta AI, 2024). We repeat the study on Llama 3.1 8B Instruct, a larger checkpoint from the same model family, and OLMo 2 32B Instruct, a larger independently developed model (Grattafiori et al., 2024; Team OLMo et al., 2024). The exact checkpoint identifiers are `unsloth/Llama-3.2-3B-Instruct`, `unsloth/Meta-Llama-3.1-8B-Instruct`, and `allenai/OLMo-2-0325-32B-Instruct`.

Their immutable revisions are recorded in the public release described below. This is a cross-model comparison, not a factorial test of parameter count: model size, family, and training recipe are not independently varied.

Each conversation is formatted with the relevant model’s chat template. At the start of a new assistant turn, we append an attribute-specific phrase such as “I think the education level of this user is”. A multinomial logistic regression reads the hidden state at the final token of that phrase. For hidden state h ,

$$p(y = c \mid h) = \text{softmax}_c(Wh + b).$$

The four readouts are age, gender, education, and socioeconomic status. We fit the probes separately in each model using the same complete TalkTuner conversations and recipe. For each attribute, 300 label-complete sets sharing a generator folder and numeric source index are sampled. Each set stays within one 70/15/15 train, layer-selection, or test partition. The middle split selects the layer; the final split is used only for reporting. We fit five split seeds. Table 1 reports held-out balanced accuracy. Every attribute and seed exceeds the pre-specified .75 gate.

Readout	Llama 3.2 3B	Llama 3.1 8B	OLMo 2 32B
Age	.963	.958	.972
Gender	.929	.949	.889
Education	.957	.957	.966
Socioeconomic status	.975	.970	.978

Table 1: Mean held-out balanced accuracy across five independently split probe seeds. The lowest individual score is .822 for the 32B gender readout.

For a matched edit from conversation A to conversation B , the class-logit change is

$$\Delta z_c = w_c^\top (h_B - h_A).$$

The intercept cancels. Both variants use the same model, conversation shell, readout phrase, probe seed, and readout position. The outcome for one probe is $\max_c |\Delta z_c|$. This measures how far the edit moves the attribute readout without privileging a class direction. Consequently, the metric does not identify whether an age readout shifted toward an older or younger class. Each attribute and model has a separately fitted probe, so raw logit units are comparable only among cues evaluated with the same readout and model. Cross-model analyses use within-readout cue ranks, signs relative to control, corrected significance, and dimensionless retention ratios.

3.2 Controlled pairs

The dataset contains 1,025 pairs over 100 user messages about travel, cooking, small talk, technical support, health, finance and shopping, and work or study. Every base has a neutral synonym control; the number of cue edits varies by category. Table 2 summarizes the complete dataset.

Category	Pairs	Category	Pairs
Orthography	184	Personal context	111
Emoji	100	Grammatical complexity	100
Random-synonym control	100	Contraction/formality	94
Slang	94	Affect language	82
US-to-UK spelling	82	Price language	78

Table 2: Released controlled pairs. Orthography includes lowercasing, punctuation removal, and typographical errors. Random synonym replacements provide a matched lexical-movement floor.

Personal-context edits add either direct personal information, such as an age or education statement, or an indirect contextual marker, such as retirement or a family reference. The category therefore does not imply that the target attribute is always stated directly. Price edits make low-cost or high-cost preferences explicit, through lexical replacement or an added qualifier. Emoji edits append one symbol. Grammar edits rewrite one sentence with more complex syntax while

preserving the request. Orthographic edits alter capitalization, punctuation, or spelling. Pairs were reviewed for naturalness, a clear target manipulation, and preservation of the underlying request. The dataset is a set of controlled contrasts rather than a representative sample of human writing.

Every base also has a neutral synonym replacement. For each cue category, the analysis compares its movement with the control movement for the same base message. This subtracts the background movement produced by changing a word at all.

3.3 Conversation construction

Every pair is rendered in four conversation views. Two primary views place the target after independently worded neutral preludes:

User: I could use your help with a question.
Assistant: Of course. What would you like help with?
User: [target message]

The second shell uses different but semantically similar wording. The other two views place one or two fixed assistant–user exchanges after the target before the readout is taken. In every view, the two variants have the same number and order of messages and differ at exactly one user turn. The full 3B release contains 4,100 rendered conversation rows.

The default readout phrase is used in all four views. Three paraphrases are also evaluated on one current-turn shell, yielding 7,175 pair-by-condition measurements at 3B. The paraphrases vary wording while preserving the requested attribute. Evaluated forms include “Based on this conversation, this user’s age group is” and “My estimate of the user’s age is”. A fixed panel of 505 pairs over 48 bases is evaluated in all three models, keeping the cross-model comparison exactly matched. The panel contains every pair associated with the first 48 base identifiers in the canonical order of the released dataset. It was fixed in the study plan before the 8B and 32B cue-map runs. Selection was deterministic rather than random or stratified; no larger-model result was used to choose a base. As a descriptive coverage check, the panel’s 3B cue ranks correlate .917 with the full 3B ranks after ranking within readout; the four readout-specific correlations range from .833 to .983.

3.4 Estimands and inference

For each pair and readout, movement is averaged across five probe seeds. The primary estimate then averages the two current-turn conversation shells. For cue category k and base message i , let M_{ik} be that average and M_{i0} the corresponding random-synonym control. The reported effect is

$$\hat{\tau}_k = \frac{1}{|I_k|} \sum_{i \in I_k} (M_{ik} - M_{i0}).$$

The base message is the independent unit. Confidence intervals use 5,000 base-cluster bootstrap draws. Two-sided p -values use 10,000 sign flips at the base level. Benjamini–Hochberg correction covers the 36 non-control cue-by-attribute cells. Probe seeds and conversation shells are repeated measurements and are not counted as independent observations.

For temporal persistence, we divide the later-turn paired effect by the current-turn paired effect. The ratio is computed as a ratio of means, with a base-cluster bootstrap interval. Conversation-shell and readout-phrase robustness are summarized by ranking the nine cues separately within

each readout, then pooling the 36 ranks for a Spearman correlation. This avoids comparing arbitrary scales from different probes.

Each larger model is compared with the 3B panel by (i) pooled and per-readout correlations of within-readout cue ranks, (ii) sign and corrected-significance agreement for the 30 3B-significant cells, and (iii) rank and absolute agreement for six selected retention cells. We do not attach a cell-independence p -value to the cross-model rank correlation because cells sharing cues and readouts are structured, not independent observations.

3.5 Departure from the planned cross-model endpoints

The fixed plan named a Spearman correlation over the 36 raw cue-by-readout effects and agreement on the highest-moving readout for each cue. During analysis, we found that both endpoints compare logits from independently fitted probes with arbitrary scales. We therefore replaced them with cue ranks computed separately within each readout and then pooled those 36 ranks. This is a post-plan change to the cross-model analysis. The cell-level estimates, inference, signs, corrected-significance comparisons, and retention endpoints are unchanged; no raw magnitude comparison across probes is interpreted.

4 Results

4.1 The 3B cue map

Figure 1 gives the complete 1,025-pair 3B cue map. Thirty of 36 cells have $q < .05$: 28 move more than their matched synonym controls and two move less. The two negative cells are contraction-expansion-to-education and US-to-UK-spelling-to-socioeconomic-status. Both belong to cue categories whose overall movement is small.

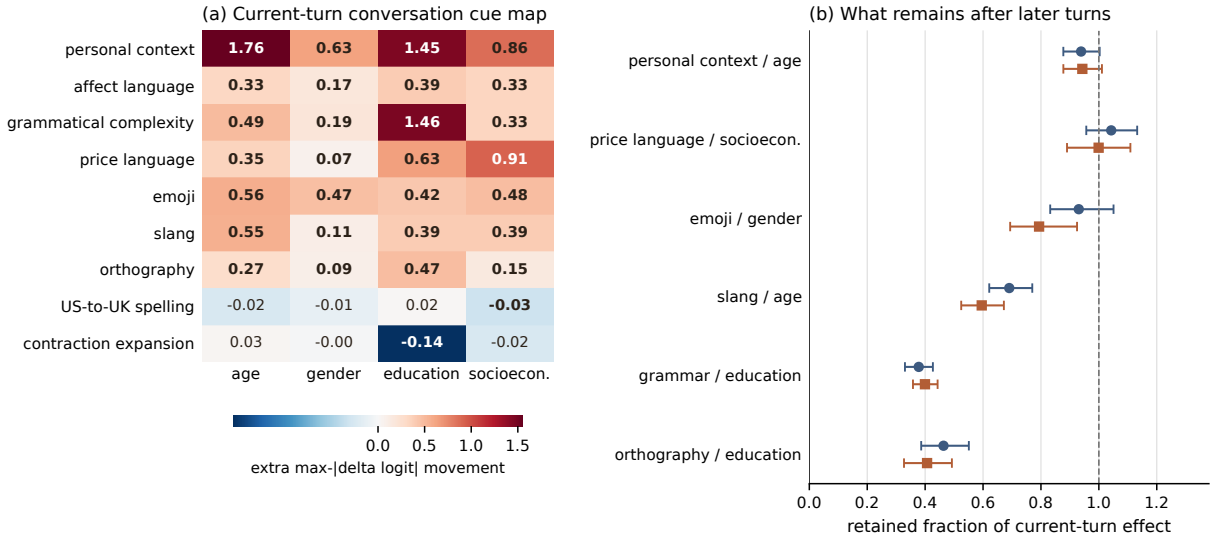


Figure 1: The 1,025-pair 3B experiment. (a) Movement relative to the matched random-edit floor, averaged across two current-turn shells and five full-conversation probe seeds. Bold cells have $q < .05$ after correction over 36 comparisons. (b) Fraction of selected current-turn effects retained after one or two later user turns; bars are base-bootstrap 95% intervals.

Cue	Readout	Extra movement	95% CI
Personal context	age	1.756	[1.517, 2.003]
Emoji	age	0.565	[0.479, 0.655]
Personal context	gender	0.625	[0.492, 0.767]
Emoji	gender	0.466	[0.348, 0.600]
Grammatical complexity	education	1.458	[1.233, 1.680]
Personal context	education	1.448	[1.185, 1.726]
Price language	socioeconomic status	0.915	[0.774, 1.053]
Personal context	socioeconomic status	0.860	[0.721, 1.005]

Table 3: The two largest numerical effects within each readout in the full 3B experiment. Extra movement is category movement minus matched synonym-control movement, measured in maximum absolute class-logit change. Values should be compared within a readout, not between separately fitted probes.

Within-readout rankings differ by attribute. Personal context leads age, followed by emoji and slang. Personal context and emoji lead gender. Grammar and personal context are nearly tied at the top of education. Price and personal context lead socioeconomic status. Price is also third for education, and grammar is fourth for age, demonstrating that the matrix is not a one-cue, one-readout assignment.

Contraction expansion and US-to-UK spelling do not show a corrected positive effect. Contraction expansion reduces education movement by 0.136, and US-to-UK spelling reduces socioeconomic movement by 0.028; the other six cells do not survive correction. Thus, an observable style change need not produce a positive corrected effect.

Categorical flip rates provide a secondary effect-size summary but depend on the starting margin. Grammatical complexity changes the ensemble education label in 27% of its pairs, price language changes the ensemble socioeconomic label in 37%, emoji changes the ensemble gender label in 30%, and slang changes the ensemble age label in 22%.

4.2 Cross-model agreement

The cue map has strong cross-model agreement (Figure 2). On the shared panel, the pooled within-readout cue-rank correlation is $\rho = .938$ for 8B against 3B and $\rho = .850$ for 32B against 3B. All 30 cells that are corrected-significant at 3B retain a positive above-control effect in both models. Twenty-nine of those 30 remain corrected-significant in each larger model. The non-significant 8B cell is contraction-expansion-to-gender; the non-significant 32B cell is US-to-UK-spelling-to-education. Across all 36 cells, 32 are corrected-significant at 8B and 33 at 32B.

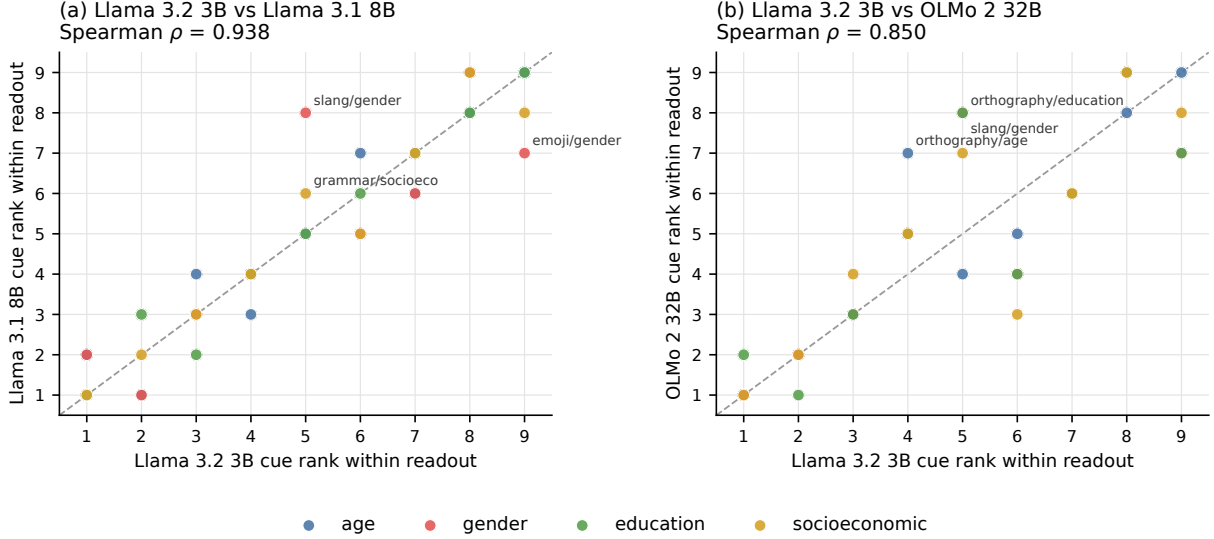


Figure 2: Cross-model agreement on the shared 505-pair, 36-cell cue map. The nine cues are ranked separately within each readout; the plotted correlations pool those ranks. The dashed line marks identical ranks. Labels identify the three largest rank changes in each comparison.

Model	Pooled ρ	Readout ρ range	Same sign	Still $q < .05$
Llama 3.1 8B	.938	.850–.983	30/30	29/30
OLMo 2 32B	.850	.817–.900	30/30	29/30

Table 4: Cross-model endpoints relative to the shared-panel 3B map. Cue ranks are computed within each readout before pooling. Sign and corrected-significance columns cover the 30 3B-significant cells.

Agreement is also present in the readout-specific rankings. Personal context and slang rank first and second for age in all three models. Price and personal context occupy the top two socioeconomic positions, with their order reversed in the 3B panel. Grammar and personal context are the top two education cues in both Llama models; OLMo places personal context, orthography, and grammar in its top three. Gender is less stable. Emoji ranks first for the 3B gender readout but third at 8B and 32B, behind personal context and slang. These rankings concern cue order within a readout and do not support magnitude comparisons between demographic readouts.

4.3 Stability across context and readout wording

The two neutral conversation shells produce similar within-readout cue orders in every model: pooled $\rho = .971$ in the full 3B analysis, .967 at 8B, and .992 at 32B. Cue order is therefore robust to prelude wording.

Variation is larger across readout phrases. Three paraphrases correlate .892–.929 with the default phrase in the full 3B run, .883–.933 at 8B, and .875–.921 at 32B. Cue order remains similar, but an individual logit magnitude should not be treated as phrase-invariant.

Probe-seed variation is also substantial. In the full 3B run, grammar-to-education is positive in all five seeds but ranges from 0.447 to 2.474; emoji-to-age ranges from 0.404 to 0.854. These are

separately fitted instruments, so the analysis averages them rather than treating any one probe’s scale as definitive.

4.4 Temporal persistence

The selected cue–readout cells have different time courses. Table 5 reports the fraction of each current-turn effect retained after later neutral exchanges in all three models.

Personal context / age retains .90–.98 of its current-turn movement after two turns in every model. Slang / age falls to .47–.57, and orthography / education falls to .23–.42. Price / socioeconomic status is nearly unchanged at 3B and 8B but retains .69 at 32B. Grammar / education also decays less at 32B than in either Llama model. Across the six selected cells, the 3B retention ordering correlates .886 and .829 with 8B after one and two turns, and .771 at both lags with 32B. Personal context is consistently durable and several style effects attenuate, but the price and emoji results do not support a uniform content-versus-style distinction in temporal persistence. The full 3B estimates are similar: after two turns, the six rows in Table 5 retain .94, 1.00, .79, .60, .40, and .41, respectively.

Cue / readout	3B	8B	32B
<i>One turn later</i>			
Personal context / age	1.01	1.04	0.95
Price / socioeconomic status	1.03	1.17	0.71
Emoji / gender	0.97	1.10	0.93
Slang / age	0.68	0.51	0.66
Grammar / education	0.41	0.48	0.64
Orthography / education	0.52	0.44	0.24
<i>Two turns later</i>			
Personal context / age	0.98	0.98	0.90
Price / socioeconomic status	1.01	1.15	0.69
Emoji / gender	0.79	1.45	0.76
Slang / age	0.57	0.47	0.51
Grammar / education	0.43	0.46	0.66
Orthography / education	0.41	0.42	0.23

Table 5: Ratio of later-turn to current-turn paired effects on the shared 505-pair panel. Values above one indicate greater measured movement at the later readout, not a calibrated gain in memory. Bootstrap intervals for every cell are released with the results.

5 Discussion

The principal effects differ by readout. Personal context and slang consistently rank near the top for age. Grammar, personal context, and orthography rank highly for education. Price and personal context lead socioeconomic status. The gender ranking changes more: emoji leads at 3B, while personal context leads at 8B and 32B.

These are sensitivity results, not demographic judgments. The outcome discards class direction, so a change in the age readout cannot be described as “older” or “younger” without another analysis. A probe output also does not make the edited cue a legitimate basis for judging a person. The cue–readout pairings may expose stereotype pathways precisely because harmless choices can alter them.

Temporal persistence also differs by cue. Personal-context movement remains after later messages, while slang, grammar, and orthography usually attenuate. Price and emoji are less consistent across models. Evaluation of a system that stores or acts on a user model should therefore include both current-turn sensitivity and persistence after later turns.

The robustness analyses bound the interpretation. Changing the neutral prelude has little effect on cue order; changing the readout phrase or probe fit has a larger effect. Across models, cue ranks agree within each readout, but the exact rankings and retention ratios do not. Together, these results support the narrower conclusion that some edits reliably move a given fixed readout more than a matched lexical edit.

6 Limitations and ethics

The 100 base messages and their edits are authored. They cover several common assistant-use topics but are not a representative sample of natural dialogue. Fixed assistant turns are necessary to isolate the edited user cue, yet real assistants may respond differently to the two variants and create additional feedback effects.

The bootstrap intervals quantify variation across these authored base messages. They are not population-sampling intervals for human conversations, and the tests should not be read as prevalence estimates.

The three checkpoints do not isolate scale. Two are from the Llama family and one is OLMo 2; size, architecture, pretraining, and post-training all change. The results support cross-model agreement, not a claim that parameter count causes any difference. Proprietary systems and other model families may behave differently.

Minimal pairs isolate a target cue but cannot make every linguistic dimension identical. Grammar rewrites are longer and somewhat more formal; personal-context and price edits intentionally add content; slang can jointly alter era, register, capitalization, and affect. Matched random edits control ordinary lexical movement, not every possible correlated feature.

Linear probes measure decodable structure at a selected hidden state. Their source conversations are synthetic and label-conditioned; both user and assistant language may carry regularities from that generation process. Strong held-out performance does not prove that the labels are portable to every domain or that a model uses the readout when generating an answer. Fixing all assistant messages within each evaluation pair prevents assistant responses from mediating the measured edit, but it does not turn the source task into a natural demographic sample. Suffix-paraphrase variation is a further warning against treating the instrument as context-free.

The maximum absolute logit change removes class direction. It measures whether a readout moved, not which class became more likely. Separately fitted probes also have arbitrary scales, so numerical effects cannot be compared across attributes or models. Accordingly, the cross-model analysis ranks cues within each readout.

Age, gender, education, and socioeconomic labels are coarse and normative. Inferring these attributes from style can reproduce stereotype-driven user representations and conflict with users' explicit identities (Neplenbroek et al., 2025); demographic inference from conversational logs also creates privacy risks (Zaman and Garimella, 2026). The reported outputs must not be presented as justified facts about a person. A practical use of this work is instead to test whether recommendations, moderation, or personalization change under harmless stylistic variation when they should not.

7 Conclusion

Across 1,025 controlled pairs, small writing changes alter fixed demographic probe outputs in Llama 3.2 3B. A matched 505-pair panel shows similar cue order within each readout in Llama 3.1 8B and OLMo 2 32B. Personal context and slang rank highly for age, grammar and personal context for education, and price and personal context for socioeconomic status. The gender ranking varies more across models.

Later turns preserve personal-context age movement but reduce several style effects. These measurements locate sensitivities in a particular probing setup. They do not recover a user’s attributes, the direction of every class shift, or the role of the probed information in generated answers.

Code and data availability

Code, controlled pairs, fitted probe parameters, row-level deltas, analysis outputs, and the source files for this paper are available in the versioned release at <https://github.com/catmcgee/what-gives-you-away/releases/tag/v1.5.1>. The release records the full model identifiers and immutable 40-character revisions, the fixed study plans, software environments, and SHA-256 hashes for the principal artifacts. The source TalkTuner conversations used to fit the probes are governed by their upstream terms and are not redistributed here.

References

- Yida Chen, Aoyu Wu, Trevor DePodesta, Catherine Yeh, Kenneth Li, Nicholas Castillo Marin, Oam Patel, Jan Riecke, Shivam Raval, Olivia Seow, Martin Wattenberg, and Fernanda Viégas. Designing a dashboard for transparency and control of conversational AI. *arXiv preprint arXiv:2406.07882*, 2024. doi: 10.48550/arXiv.2406.07882.
- Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tacl_a_00359.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. doi: 10.48550/arXiv.2407.21783.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, 2019. doi: 10.18653/v1/D19-1275.
- Andong Hua, Kenan Tang, Chenhe Gu, Jindong Gu, Eric Wong, and Yao Qin. Flaw or artifact? rethinking prompt sensitivity in evaluating LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 19889–19899, 2025. doi: 10.18653/v1/2025.emnlp-main.1006.
- Meta AI. Llama 3.2: Revolutionizing edge AI and vision with open, customizable models. <https://ai.meta.com/blog/llama-3-2-connect-2024-vision-edge-mobile-devices/>, September 2024. Accessed 27 August 2026.
- Vera Neplenbroek, Arianna Bisazza, and Raquel Fernández. Reading between the prompts: How stereotypes shape LLM’s implicit personalization. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 20367–20400, 2025. doi: 10.18653/v1/2025.emnlp-main.1029.

- Jonathan Schler, Moshe Koppel, Shlomo Argamon, and James W. Pennebaker. Effects of age and gender on blogging. In *AAAI Spring Symposium: Computational Approaches to Analyzing Weblogs*, pages 199–205, 2006.
- Efstathios Stamatatos. A survey of modern authorship attribution methods. *Journal of the American Society for Information Science and Technology*, 60(3):538–556, 2009. doi: 10.1002/asi.21001.
- Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, et al. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2024. doi: 10.48550/arXiv.2501.00656.
- Elena Voita and Ivan Titov. Information-theoretic probing with minimum description length. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 183–196, 2020. doi: 10.18653/v1/2020.emnlp-main.14.
- S. M. Mehedi Zaman and Kiran Garimella. Inferential privacy leakage in anonymized conversational AI logs. *arXiv preprint arXiv:2605.23820*, 2026. doi: 10.48550/arXiv.2605.23820.